



Department of Electrical Engineering  
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**MT-302 Probability and Statistics**  
**Assignment # 02**

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Submitted To:

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# Probability Questions

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1.

scratch resistance

|   |      | <u>shock resistance</u> |     |
|---|------|-------------------------|-----|
|   |      | high                    | Low |
| B | high | 40                      | 4   |
|   | Low  | 1                       | 5   |
|   |      | A                       |     |

$A \cap B$ :- 40

$A \cup B$ :-  $1 + 40 + 4 = 45$

$A'$ :-  $5 + 4 = 9$

2.

supplier

1  
2  
3

|   |    | <u>conforms</u> |    |
|---|----|-----------------|----|
|   |    | yes             | no |
| A | 22 | 22              | 8  |
|   | 25 | 25              | 5  |
|   | 30 | 30              | 10 |
|   |    | B               |    |

(a)  $P(A)$

A:-  $= \frac{22 + 8}{100} = 0.3.$

(b)  $P(B)$

A:-  $= \frac{22 + 25 + 30}{100} = 0.77.$

(c)  $P(A')$

A:-  $P(A') = \frac{25 + 5 + 30 + 10}{100} = 0.7.$

(d)  $P(A \cap B)$

A:-  $= \frac{22}{100} = 0.22.$

(e)  $P(A \cup B)$

A:-  $= \frac{85}{100} = 0.85.$

(f)  $P(A' \cup B)$

A:-  $= \frac{92}{100} = 0.92.$

3.

surface finish

|           |   | <u>Edge finish</u> |      |
|-----------|---|--------------------|------|
|           |   | Excellent          | Good |
| Excellent | A | 80                 | 2    |
|           | B | 10                 | 8    |

$$(a) P(A) = 82/100 = 0.82$$

$$(b) P(B) = 90/100 = 0.9$$

$$(c) P(A') = 18/100 = 0.18$$

$$(e) P(A \cup B) = 92/100 = 0.92$$

$$(d) P(A \cap B) = 86/100 = 0.8$$

$$(f) P(A' \cup B) = 98/100 = 0.98$$

4.

scratch resistance

shock resistance

|      |      |     |   |
|------|------|-----|---|
|      | High | Low |   |
| High | 70   | 9   | A |
| Low  | 16   | 5   | B |

$$(a) P(A \cap B) = 70/100 = 0.7$$

$$(b) P(A \cup B) = 95/100 = 0.95$$

(c) As  $(A \cap B) \neq \phi$  so it is not mutually exclusive.

5.

conductivity

Strength

|      |      |     |   |
|------|------|-----|---|
|      | High | Low |   |
| High | 71   | 15  | A |
| Low  | 11   | 3   | B |

$$(a) P(A \cap B) = 71/100 = 0.71$$

$$(b) P(C \cup D) = 29/100 = 0.29$$

(c) Both events are not mutually exclusive.

6.

surface finish conforms

shock Resistance  
roundness conforms

|     |     |    |   |
|-----|-----|----|---|
|     | yes | no |   |
| yes | 70  | 9  | A |
| no  | 16  | 5  | B |



- (a)  $P(A) = 79/100 = 0.79$   
 (b)  $P(A \cup B) = 95/100 = 0.95$   
 (c)  $P(A \cup C) = 84/100 = 0.84$   
 (d)  $P(A \cap B) = 70/100 = 0.7$

7. Final Temperature conditions

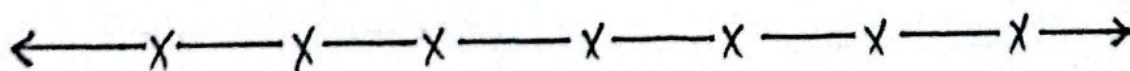
Heat absorbed (cal)

|       | Below Target | Above Target |
|-------|--------------|--------------|
| 266 K | 12           | 40           |
| 271 K | 44           | 16           |
| 274 K | 56           | 36           |

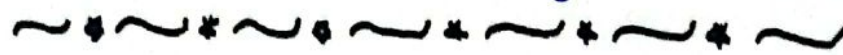
$$P(A) = 0.549, \quad P(B) = 0.451$$

$$\begin{aligned} \text{(a) } P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.549 + 0.451 - \left( \frac{40 + 16}{204} \right) = 0.725 \end{aligned}$$

$$\begin{aligned} \text{(b) } P(A' \cup B') &= 1 - P(A \cup B) \\ &= 1 - 0.725 = 0.275 \end{aligned}$$



# Conditional Probability Questions



2-57.

scratch  
resistance

shock Resistance

|      | high | low |   |
|------|------|-----|---|
| high | 70   | 9   | B |
| Low  | 16   | 5   |   |
|      | A    |     |   |

$$(a) P(A) = \frac{70+16}{100} = 0.86$$

$$(b) P(B) = \frac{70+9}{100} = 0.79$$

$$(c) P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{70/100}{0.79} = 0.886$$

$$(d) P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{16/100}{0.86} = 0.184$$

2-58.

surface finish

length

|           | excellent | good |   |
|-----------|-----------|------|---|
| excellent | 80        | 2    | A |
| good      | 10        | 8    | B |
|           |           |      | C |

$$(a) P(A) = 82/100 = 0.82$$

$$(b) P(B) = 90/100 = 0.9$$

$$(c) P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{80/100}{0.9} = 0.889$$

$$(d) P(B|A) = \frac{80/100}{0.82} = 0.976$$

$$(e) P(B|A) = 0.976$$

$$(f) P(A|C) = \frac{P(A \cap C)}{P(C)} = \frac{2/100}{10/100} = 0.2$$

2-59.

surface finish  
conforms

roundness conforms

|       |     |     |
|-------|-----|-----|
|       | yes | no  |
| yes A | 345 | 5   |
| no B  | 12  | 8 C |

$$(a) P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{345/375}{\frac{345+12}{375}} = 0.966$$

$$(b) P(A|C) = \frac{5/375}{\frac{5+8}{375}} = 0.385$$

2-60.

surface roughness

coating weight

|        |      |      |
|--------|------|------|
|        | high | low  |
| high B | 12   | 16 D |
| low A  | 88   | 34 C |

$$(a) P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{12/150}{\frac{12+88}{150}} = 0.12$$

$$(b) P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{12}{12+16} = 0.015$$

$$(c) P(D|C) = \frac{P(D \cap C)}{P(C)} = \frac{34}{88+34} = 0.184$$

2-61 (excluded)

2-62

(a) On 1<sup>st</sup> draw:-

$$P(\text{defective 1st}) = \frac{20}{100} = 0.2$$

(b) 2<sup>nd</sup> draw:-

$$P(2^{\text{nd}} \text{ defective}) = \frac{19}{99} = 0.192$$

$$\begin{aligned} (c) \quad P(\text{both defective}) &= P(1^{\text{st}} \text{ defective}) \times P(2^{\text{nd}} \text{ defective}) \\ &= 0.2 \times 0.192 \\ &= 0.038 \end{aligned}$$

